Chi-Square Tests, Pt 1

CSU, Chico Math 351

2026-04-13

outline

outline

Introduction

This section discusses ways to test if data come from a particular distribution. The observed data should be close to what we would expect under the assumption that the data belong to a particular distribution. In this section we consider what we call **categorical** data. This implies that each observation can be classified into a number of possible categories or types. In particular, we consider:

1. Chi-Square Goodness of Fit: Testing whether a categorical variable follows a particular distribution.
2. Chi-Square Test of Independence: Testing whether two categorical variables are independent.

outline

Definition

Consider the binomial model. Suppose that X_1 follows a binomial distribution with parameter n and p_1 where $0 < p_1 < 1$. According to the CLT,

$$Z = \frac{X_1 - np_1}{\sqrt{np_1(1-p_1)}}$$

has a standard Normal distribution as long as n is large ($np_1 > 5$ and $n(1-p_1) > 5$).

Definition

Now consider the distribution of $D_1 = Z^2$:
If we let $X_2 = n - X_1$ and $p_2 = 1 - p_1$, we can write D_1 as:

$$D_1 = \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_1 - np_1)^2}{n(1 - p_1)}$$

Since

$$(X_1 - np_1)^2 = (n - X_1 - n[1 - p_1])^2 = (X_2 - np_2)^2$$

giving us

$$D_1 = \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_2 - np_2)^2}{np_2}$$

Definition

Just as we've seen with past tests, our test statistic here will consider the difference between observed and expected values, adjusted for some measure of error.

Since we know that $n = X_1 + X_2$ and $1 = p_1 + p_2$, we are effectively comparing the number of successes observed (X_1) to those expected $(n * p_1)$, and the number of failures observed (X_2) to those expected $(n * p_2)$.

D_1 follows a χ^2 Distribution. If the observed and expected values are equal, $D_1 = 0$, and the further apart they are, the larger D_1 will become.

Expanding the Definition

Success and failure are mutually exclusive (and exhaustive) outcomes, so in this case, we have two categories. But we can extend this for any number k of mutually exclusive categories.

Suppose we have $A_1, ..., A_k$ mutually exclusive and exhaustive outcomes, each of which have $\Pr(A_i) = p_i$ and thus $\sum_{i=1}^{k} p_i = 1$. Suppose that an experiment is repeated n times and we let X_i represent the number of times the experiment results in A_i, $i = 1, ..., k$. The joint distribution of $X_1, X_2, ..., X_{k-1}$ is

$$f(x_1, x_2, ..., x_{k-1}) = \Pr(X_1 = x_1, X_2 = x_2, ..., X_{k-1} = x_{k-1})$$

where $x_1, ..., x_{k-1}$ are nonnegative integers such that $x_1 + \cdots + x_{k-1} \leq n$.

Expanding the Definition

The end result: for k such groups we have

$$D = \sum_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i}$$

where D has an approximate chi-square distribution with $k - 1$ degrees of freedom.

Even more generally:

$$D = \sum_{i=1}^{k} \frac{(X_i - E[X_i])^2}{E[X_i]} = \sum_{i=1}^{k} \frac{(Observed_i - Expected_i)^2}{Expected_i}$$

P-value reminder for R: For the probability of observing values as far or further away from what is expected, we can, like before, use **1-pchisq(D, k-1)**, where D is the test statistic and k is the number of categories in the variable.

outline

Example 1 (from Larsen and Marx)

Ceriodaphnia cornuta, a crustacean, has "horned" and "unhorned" variants. To test whether they are equally likely to be eaten, a large number of the crustacean were added to a holding tank in a 3-to-1 ratio of unhorned to horned. After analyzing how many were consumed in the given time period, they found the ratio of unhorned-to-horned consumed was forty to four. What do these counts imply? Let $\alpha = 0.05$

For this distribution, we would **expect**, if $p_1 =$ unhorned eaten and $p_2 =$ horned eaten, that:

$$H_0 : p_1 = 3/4, p_2 = 1/4$$

$$H_1 : p_1 \neq 3/4, p_2 \neq 1/4$$

Example 1 (from Larsen and Marx)

Given the "consumed" ratio, we can define the sample's test statistic d:

$$d = \frac{[40 - 44 * (3/4)]^2}{44 * 3/4} + \frac{[4 - 44 * (1/4)]^2}{44 * 1/4} \approx 5.94$$

We can compare against a critical value to see if we reject:

chi.critical = qchisq(1-alpha,1) = 3.841

And we can calculate the p-value itself by using:

p-value = 1-pchisq(5.94,1) = 0.01489

In either case, we confirm we **reject** the null hypothesis. It appears the shape of the crustacean does affect its chances of being eaten.

Exercise 1 (from Speegle and Clair)

Diaconis, Holmes and Montgomery claim that vigorously flipped coins tend to come up the same way they started. In a real coin tossing experiment, two UC Berkeley students tossed coins a total of 40 thousand times in order to assess whether this is true. Out of the 40,000 tosses, 20,245 landed on the same side as they were tossed from. Use a hypothesis test with $\alpha = 0.05$ to test the claim.

1. Clearly state the null and alternative hypotheses, defining any parameters that you use.
2. List the test statistic and p-value, showing your work.
3. Do we reject H_0?
4. What do we conclude about the coin tossing experiment?

outline

references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Richard J Larsen, Morris L Marx. *An Introduction to Mathematical Statistics and Its Applications.* Prentice Hall, fifth edition, 2012.

Adapted from notes by Dr. Gray.